



34 Connecting the Masses



40 The Hand That Rocks The Rig



62 Giving Back To The Machines

Digital

Passion

Fuelling The Pursuit Of Technology Knowledge

Main Story

The

Red Brick ROAD

We're on the brink of another of humanity's historic voyages. We're talking bigger than Armstrong landing on the moon, or any other achievement involving space travel: what will help humans make the trip to mars, and more importantly, get them back safely.

Robert Sovereign-Smith

Named after the Roman god of war, looking fiery red in our telescopes, the fourth rock from the sun, our first stepping stone into "outer space" and perhaps even future home for humans, Mars has always elicited wonderment from us. Until quite recently, in terms of planetary ages, we really didn't know much about the Red planet... here's a brief timeline:

In 1877, Mars' two moons were discovered by astronomer Asaph Hall, and were named Deimos (panic) and Phobos (fear)—after the horses that pulled the chariot of the Roman god of war.

In 1971, Mariner 9 orbited Mars, becoming the first space probe to orbit another planet. It took pictures of the Martian surface and discovered Mount Olympus, the largest known volcano in our solar system—27 kilometres tall (Mount Everest is a mere 8.8 kilometres high), and a whopping 550 kilometres wide!

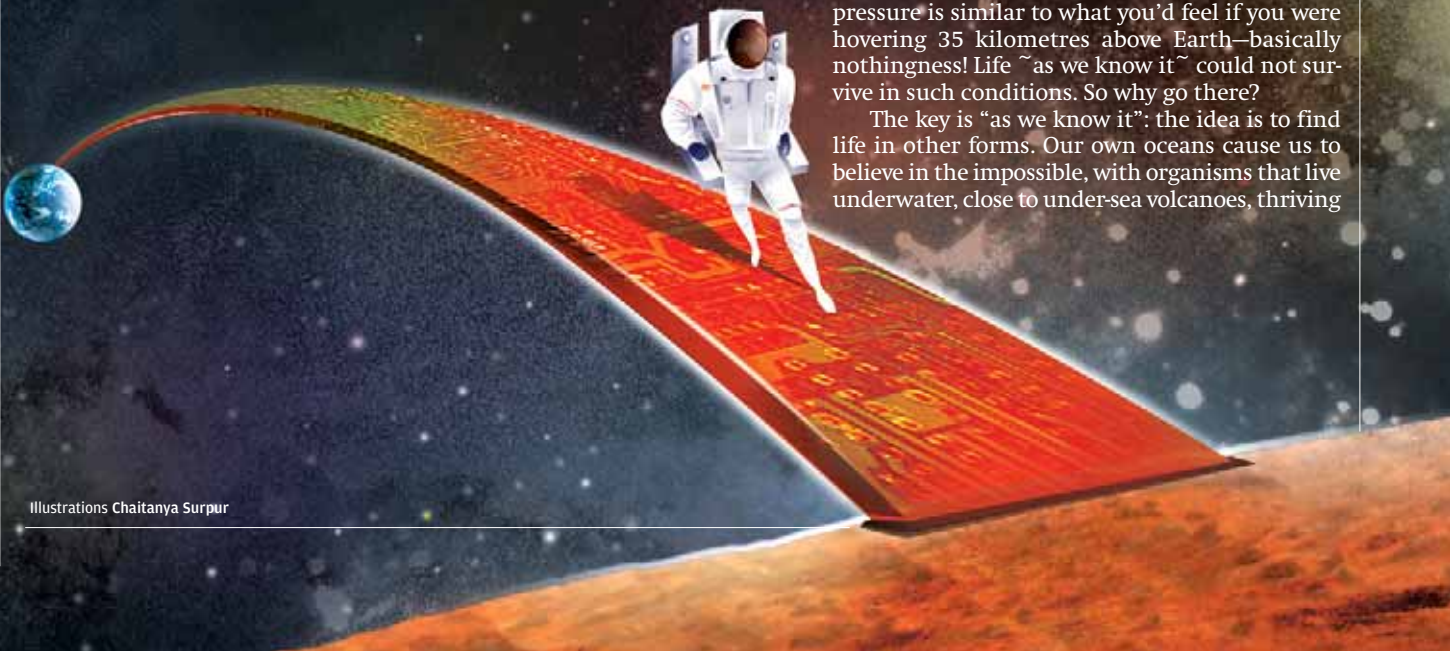
In 1975, the Viking I and II spacecraft landed on Mars. They sampled the soil and rocks and sent back vital information.

Since then, there have been numerous spacecraft that have orbited, flown by or landed on Mars, each one making new discoveries. For example, we now know that the elixir of life, good old liquid H₂O, existed on Mars at one point. We also know that there is enough ice at the poles to cover the entire Martian surface with water—if the polar caps were to be melted.

Why Should We Go There?

The conditions on Mars are hostile—400 kmph winds, icy cold, carrying tiny particles of rust. The atmosphere is sparse, made up of mostly carbon dioxide (95%), nitrogen (3%), argon (a little over 1.5%) and mere traces of oxygen. The atmospheric pressure is similar to what you'd feel if you were hovering 35 kilometres above Earth—basically nothingness! Life ~as we know it~ could not survive in such conditions. So why go there?

The key is "as we know it": the idea is to find life in other forms. Our own oceans cause us to believe in the impossible, with organisms that live underwater, close to under-sea volcanoes, thriving



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